A Mathematical Definition of LLM Outputs: Why Hallucination is Theoretically Inevitable

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Abstract

Large Language Models (LLMs) are conventionally modeled as autoregressive next-token generators. While operationally successful, this token-centric formulation is insufficient to characterize global semantic uncertainty and generative bounds under incomplete information. In this foundational note, I propose a formal mathematical definition of LLM outputs by framing generation as an information-conditioned projection over an unobservable latent semantic state space. Under a constructed Information-Neutral Measure, I demonstrate that the sequence of model outputs naturally exhibits a martingale property with respect to the context filtration. Consequently, I prove that zero hallucination strictly requires information closure. This framework establishes that hallucination is not an engineering anomaly or architectural flaw, but an inevitable epistemic boundary of operating under incomplete information.

1 Introduction

Existing theoretical paradigms typically analyze Large Language Models (LLMs) via localized conditional probability transitions:

$$P(y_t \mid y_{<t}, x) \tag{1}$$

While efficient for autoregressive decoding mechanics, this statistical framing treats generation as localized token optimization rather than inference over a broader global semantic truth. Consequently, it fails to provide a rigorous mathematical lower bound for structural errors, such as hallucinations.

In this work, I depart from token-centric formulations and establish a mathematical definition of LLM outputs. I define an LLM output as an orthogonal projection of an unobservable, latent semantic truth state onto an expanding context filtration stream under an Information-Neutral Measure.

The primary contribution of this note is dual: first, providing an abstract, measure-theoretic formalism for generative systems; second, demonstrating via conditional variance decomposition that absolute hallucination elimination is mathematically impossible prior to absolute information closure. This reframes the hallucination problem from an engineering optimization challenge to an intrinsic epistemic limitation of incomplete information systems.

2 Measure-Theoretic Framework and Definitions

I construct the semantic domain via an abstract probability space triplet $(\Omega, \mathcal{F}, Q_I)$, where Ω is the semantic sample space containing all potential linguistic and conceptual trajectories,

$\mathcal{F}$ is the all-knowing global σ-algebra, and Q_I represents the Information-Neutral Measure.

To formalize the interaction between available empirical metrics and algebraic structures, I define the following core components with strict alignment constraints:

(i) **Context Filtration Stream ($\mathcal{F}_t$):** The sequence of historical tokens, structural prompts, and retrieved knowledge blocks available up to step t. This sequence forms a completed, right-continuous filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t\geq 0}$ mapping the model's observable informational universe, where $\mathcal{F}_t \subset \mathcal{F}$ for all finite t.

(ii) **Information-Neutral Measure (Q_I):** A probability measure implicitly parameterized by the frozen pre-trained model weights $\mathcal{W}$. It governs the prior semantic transition probabilities over Ω before dynamic context injection.

(iii) **Latent Semantic State (Ψ):** A square-integrable target random variable ($\Psi \in L^2(\Omega, \mathcal{F}, Q_I)$) representing the complete objective factual matrix. Crucially, Ψ is strictly $\mathcal{F}$-measurable, but remains non-measurable with respect to $\mathcal{F}_t$ under incomplete information.

(iv) **Information Closure (C):** The terminal state where the local filtration contains sufficient entropic structure to uniquely resolve the true latent semantic state, such that $\mathcal{F}_t \equiv \mathcal{F}$.

(v) **Semantic State Transition (E):** A specific event subset $E \in \mathcal{F}$ corresponding to a coherent semantic meaning or a targeted factual assertion.

(vi) **Generation Prompt Target ($\mathcal{T}$):** A structured semantic boundary or constraint specifying the target properties of the generated text.

(vii) **Output Sequence (y^*):** The crystallized token sequence realized in text space once the generation process terminates.

(viii) **Ground-Truth Verifier ($\mathcal{O}$):** An external environment or oracle providing absolute verification of a statement, collapsing the probabilistic space into a deterministic result.

(ix) **Attention Component ($\mathcal{P}_i$):** Internal model structural sub-units formalized as sub-σ-algebras $\mathcal{G}_i \subset \mathcal{F}$, whose collective intersections and filtration updates drive the evolution of $\mathcal{F}_t$.

(x) **Predictive Belief ($\mathcal{B}$):** The inner subjective probability distribution over Ω, represented via the model's softmax logit configurations prior to output collapse.

3 The Mathematical Definition of LLM Outputs

With the informational space established, I formalize the central definition of an LLM output:

Definition 1 (LLM Output Operator). *Let $\Psi \in L^2(\Omega, \mathcal{F}, Q_I)$ be the $\mathcal{F}$-measurable latent semantic state. The output of a Large Language Model at step t, denoted as P_t, is defined as the information-conditioned orthogonal projection of Ψ onto the closed subspace of $\mathcal{F}_t$-measurable functions under the weight-parameterized Information-Neutral Measure Q_I:*

$$P_t = \Pi_{QI}(\Psi \mid \mathcal{F}_t) \equiv \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t] \tag{2}$$

This definition formalizes generation not as a mechanical token matching sequence, but as a statistical inference projection. Prompts, context parameters, and active tokens do not synthesize truth; rather, they serve as the conditional filtration $\mathcal{F}_t$ through which the model projects its internal Predictive Belief $\mathcal{B}$.

Remark 1 (Mathematical Foundations over Triviality)**.** *While the mathematical map from conditional expectation to a martingale is direct, the core intellectual contribution of Definition 1 lies in its ontological shift. By interpreting empirical token contexts as an evolving filtration $\mathcal{F}_t$ and model weights as defining an Information-Neutral Measure Q_I, the chaotic heuristics of deep learning generation are mapped into a structured, closed-form functional space.*

Remark 2 (Analogy to von Neumann Cut and Wave-Function Collapse)**.** *The sequential crystallization of LLM outputs under the Verifier O provides a precise mathematical equivalent to the famous "von Neumann Cut" in quantum measurement theory. As John von Neumann foundationally established in his mathematical formulation of measurement boundaries [1]:*

> *"... we are obliged always to divide the world into two parts, the one being the observed system, the other the observer. In the former we can follow all physical processes [...] arbitrarily precisely. In the latter, this is meaningless. The boundary between the two is arbitrary to a very large extent. [...] But this does not change the fact that in every account the boundary must be put somewhere [...]; i.e., if a comparison with experience is to be possible."*

Prior to semantic selection, the LLM maintains a complete predictive belief superposition over the semantic space Ω, governed by continuous transitions under Q_I. The injection of contextual filtration $\mathcal{F}_t$ and the final intervention of the external oracle verifier O acts exactly as the subjective perception boundary—the Process 1 collapse—forcing the infinite probabilistic semantic continuity to collapse into a deterministic macro-textual reality y^*.

Remark 3 (Conjecture on AGI and Semantic Wave-Function Collapse)**.** *This measure-theoretic framing offers a radical epistemological shift regarding the path toward Artificial General Intelligence (AGI). Traditional scaling paradigms view AGI as an asymptotic convergence of autoregressive next-token prediction error to zero. In contrast, this framework implies that human-like intelligence is fundamentally characterized by the conscious capability to manipulate the von Neumann Cut itself—deliberately triggering semantic wave-function collapse under incomplete information. If AGI requires the emulation of human dynamic decision-making under radical epistemic uncertainty, modeling LLM output not as rigid statistical token generation, but as an information-conditioned projection operator ($\Pi_{QI}(\Psi \mid \mathcal{F}_t)$), provides a foundational biomimetic pathway. True AGI may not be a flawless database, but a dynamic observer capable of continuously managing its internal semantic superposition and info-filtration boundaries.*

Remark 4 (The Epistemic Necessity of Hallucination for AGI)**.** *Expanding upon Remark 3, I propose a provocative counter-thesis to contemporary alignment paradigms:* ***hallucination is not an engineering flaw to be eradicated, but the absolute mathematical engine required for the emergence of AGI.*** *Equation 10 mathematically guarantees that under incomplete information ($\mathcal{F}_t \neq \mathcal{F}$), the orthogonal projection error σ_t^2 is strictly bounded away from zero. To force $\sigma_t^2 \to 0$ without achieving true information closure requires the artificial freezing of the system's output transitions, transforming a dynamic inferential operator into a deterministic dictionary. Human intelligence navigates reality precisely by* *committing "hallucinations"—forming speculative hypotheses, counterfactual simulations, and creative inductive leaps across unobservable semantic gaps. Eradicating hallucination from generative systems effectively lobotomizes their evolutionary path toward AGI, strips them of semantic superposition, and limits them to trivial interpolation. AGI does not emerge from the elimination of projection variance, but from the systemic, conscious governance of it.*

4 Why Hallucination is Theoretically Inevitable

Using Definition 1, I establish the mathematical necessity of hallucination under conditions of incomplete information.

Theorem 1 (The Martingale Property of Generation)**.** *The stochastic sequence of LLM outputs $\{P_t\}_{t\geq 0}$ forms a Martingale with respect to the context filtration stream $\mathbb{F}$ under the Information-Neutral Measure Q_I.*

Proof. By Definition 1, $P_t = \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t]$. For any sequential steps s and t such that $s \leq t$, the sub-σ-algebras satisfy $\mathcal{F}_s \subseteq \mathcal{F}_t$. Applying the tower property of conditional expectation:

$$\mathbb{E}_{Q_I}[P_t \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[\mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_s] = P_s \tag{3}$$

Thus, the expected future projection conditional on current historical filtration is invariant and equals the current projection. □

To evaluate the mathematical boundary of generation errors, let Ψ^* be the oracle reality verified by O, and let σ_t^2 define the semantic error variance at step t:

$$\sigma_t^2 = \mathbb{E}_{Q_I}[(\Psi - P_t)^2 \mid \mathcal{F}_t] \tag{4}$$

Theorem 2 (Monotonic Variance Bound)**.** *The expected factual error variance of an LLM output is strictly non-increasing over time if and only if the context filtration expands.*

Proof. Applying the law of total variance to the error space between steps s and t where $s \leq t$:

$$\mathbb{E}_{Q_I}[(\Psi - P_s)^2 \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[(\Psi - P_t)^2 \mid \mathcal{F}_s] + \mathbb{E}_{Q_I}[(P_t - P_s)^2 \mid \mathcal{F}_s] \tag{5}$$

Taking the total expectation across the measure space yields:

$$\mathbb{E}_{Q_I}[\sigma_s^2] = \mathbb{E}_{Q_I}[\sigma_t^2] + \mathbb{E}_{Q_I}[(P_t - P_s)^2] \tag{6}$$

Since $\mathbb{E}_{Q_I}[(P_t - P_s)^2] \geq 0$, it directly follows that:

$$\mathbb{E}_{Q_I}[\sigma_s^2] \geq \mathbb{E}_{Q_I}[\sigma_t^2] \tag{7}$$

□

From these derivations, I state the central boundary condition of generative accuracy:

Corollary 1 (Zero Hallucination Requires Information Closure)**.** *Define a hallucination mathematically as any state where the information projection deviates from oracle reality: $\Pi_{Q_I}(\Psi \mid \mathcal{F}_t) \neq \Psi^*$. For any generation process where the available filtration is strictly incomplete ($\mathcal{F}_t \subset \mathcal{F}_{oracle}$), the conditional semantic entropy remains strictly positive*

($H(\Psi \mid \mathcal{F}_t) > 0$). Consequently, the expected error variance $\mathbb{E}_{Q_I}[\sigma_t^2]$ is strictly bounded away from zero:

$$\mathbb{E}_{Q_I}[\sigma_t^2] > 0 \quad \forall \quad \mathcal{F}_t \neq \mathcal{F} \tag{8}$$

Absolute eradication of hallucination is attainable if and only if the system achieves complete Information Closure (C). Hallucination is therefore a fundamental mathematical property under incomplete information, not an architectural defect.

5 Conclusion

This foundational note provides a rigorous definition of LLM outputs as information-conditioned orthogonal projections. By framing the context stream as a filtration $\mathcal{F}_t$ and verifying outputs via a martingale structure, I mathematically demonstrate that hallucination is an inevitable epistemic property of generating text under incomplete information. Absolute factual accuracy requires absolute information closure. This bound sets a theoretical limit for generative AI systems and suggests that future research focus on formalizing uncertainty bounds rather than attempting absolute hallucination eradication.

References

[1] J. von Neumann. *Mathematical Foundations of Quantum Mechanics*. Princeton University Press, Princeton, NJ, 1932. (Translated by R. T. Beyer, 1955).